Change sheets

GNSS Based Orbit Determination and Trajectory Prediction — project report (26-page printing)

These sheets contain only the pages of the printed report whose content changed: the carrier-phase orbit determination on the Orekit engine and the five further improvements with their two-month results. All other printed pages stay exactly as they are.

How to update the printed copy: for each block below, remove the printed pages listed and put the enclosed pages in their place. Each enclosed page is labelled in its top margin.

Remove printed pages	Insert enclosed pages	Content
9–11	9, 10, 11, 11a, 11b	§9 Computational methods · §10 Software architecture · §11 Processing pipeline
18	18, 18a, 18b	§13 results: §13a carrier-phase rebuild and new §13b five improvements
21–22	21, 22, 22a	§17 Limitations · §18 Running the software
26	26	Glossary (end): new terms

7 printed pages are replaced by 12 pages; the report grows from 26 to 31 pages.
Pages 1–8 (title, contents, Parts I–IV), 12–17, 19–20 and 23–25 are unchanged.
The table of contents is unchanged: the new results are subsections §13a and §13b of §13.
Where a block ends part-way down a page, the rest of that page is intentionally blank so that the next unchanged printed page follows on correctly.

- **Ap** — a daily **geomagnetic** activity index. During geomagnetic storms, energy is deposited in the upper atmosphere, heating it and raising density globally. Higher Ap implies greater drag.

These indices are read from a space-weather file (a CelesTrak CSSI file for historical values), with constant values used when none is supplied. Because F10.7 and Ap cannot be predicted precisely days in advance, even a perfect density model leaves an irreducible forecast uncertainty — the principal reason the forecast error grows to approximately 20 km over seven days (§13–§14).

Part V — The mathematics

9. Computational methods

Four computational stages convert raw GPS measurements into an evaluated forecast.

9.1 Single-point positioning. At each epoch there are several pseudoranges and four unknowns (position and clock error), generally over-determined and noisy. The position and clock that best fit all the pseudoranges are found by **least squares** — minimising the sum of squared differences between measured and computed distances. As distance depends non-linearly on position, the solution is obtained iteratively (linearise, correct, repeat). Each epoch yields one position **fix** and associated quality indicators. In the current system these fixes only *start* the orbit determination (§9.3): three of them give an initial orbit (Gibbs' method) and a quick fit to all of them refines it.

9.2 Numerical integration. The equation of motion (§7) specifies the acceleration at any state; the trajectory over time is required. No closed-form solution exists once drag and non-spherical gravity are included, so the orbit is integrated **numerically**, advancing the state in small time steps and re-evaluating every force at each step. The software uses Orekit's **Dormand–Prince 8(5,3)** method, a **Runge–Kutta** integrator that computes each step at two orders of accuracy; the difference estimates the local error, which is used to **adapt the step size** automatically. A **position tolerance** (1 mm by default) sets the permitted per-step error; a tighter tolerance produces more, smaller steps — more accurate but slower. Integrating the summed forces directly in this way is **Cowell's method**.

9.3 Batch least-squares orbit determination on carrier phase. The orbit is fitted directly to the GPS measurements, not to position fixes. For every GPS satellite in view, every 60 s over a 12-hour arc, two measurements are formed from the dual-frequency data, both free of the first-order ionospheric delay (the **ionosphere-free combination**):

- the **code** (pseudorange), noisy at about a metre;
- the **carrier phase**, the count of radio-wave cycles, about a hundred times less noisy. It carries an unknown whole number of cycles, the **ambiguity**, which is constant while

the receiver keeps lock.

The unknowns are:

- the initial orbit (position and velocity) and the **drag coefficient**;
- one **receiver clock error** per epoch;
- one **ambiguity** per continuous phase arc.

Typically that is about 800 unknowns fitted to about 10 000 measurements. For each trial orbit, Orekit propagates the trajectory (§9.2) and computes the measurements. The computation includes the signal travel time, the Earth's rotation during it, the **Shapiro** relativistic delay, the GPS clock, the positions of both antennas, and the **phase wind-up** (§13b). The **residuals** are the measured minus computed values.

The **Jacobian** (the sensitivity of every residual to every unknown) comes from Orekit's variational equations. **Gauss–Newton** iterations then reduce the weighted squared residuals until the corrections become negligible.

Between solves the data are cleaned:

- **cycle slips** (sudden jumps of whole cycles) split an arc;
- **outliers** are removed;
- code and phase are **re-weighted** by their actual scatter, so the **variance factor** (observed versus assumed noise) is near one.

The fit quality is reported as the **post-fit RMS** of the phase residuals, typically 6–7 cm.

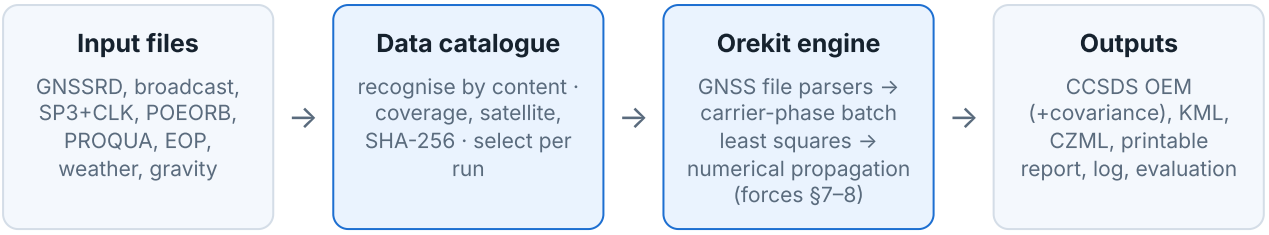
9.4 Covariance propagation. A single predicted position is incomplete without its uncertainty. The **covariance** is a matrix expressing the uncertainty of each state component and the correlations between them. Orbit determination yields a covariance at the arc start. To report it along the forecast, the software propagates it with the **state-transition matrix** (the matrix relating a small change in the initial state to the change at a later time) and with the sensitivity to the estimated drag coefficient. Both are integrated by Orekit alongside the orbit. The result is written into the OEM and can be displayed as an uncertainty ellipsoid or as radial/along/cross (RTN) standard-deviation curves. Because covariance costs approximately one Covariance propagation costs extra computation, so it is optional and disabled in the fast default profile.

Part VI — System design

10. Software architecture

The system is a single Java application. Its flight-dynamics engine is **Orekit**, the open-source space flight-dynamics library (Apache 2.0). Orekit is packed into the same

executable file, so the system still runs offline. It is organised in three layers.



A local web server on 127.0.0.1 serves the browser interface (run setup, 3-D globe, 2-D map, plots, comparisons) over the engine; no data leaves the computer.

Layer 1 — the engine (Orekit). Every run executes on Orekit classes:

- **Data:** a fresh Orekit data context per run, holding the leap seconds, Earth orientation, gravity field, ocean tides, DE440 ephemeris and the GPS antenna file.
- **Input files:** the RINEX observation, SP3, clock and navigation parsers.
- **Physics:** the IERS 2010 frames and time scales, the force models of Part IV, and the attitude laws.
- **Orbit determination:** the batch least-squares estimator with the GNSS code and phase measurement models.
- **Forecast:** the numerical propagator and its covariance.

The program adds its own logic around Orekit:

- the product-tier and cut-off rules (§6);
- the single-point positions that start the fit;
- data editing (cycle slips, outliers);
- the cut-off-safe Earth-orientation and space-weather inputs;
- a solver that exploits the sparse structure of the problem;
- the OEM, KML and CZML writers.

Representative packages: `orekit` (the engine), `pipeline` (run configuration and workflows), `gnss` (GNSS data handling), `data` (the catalogue) and `io` (file formats).

Layer 2 — the data catalogue. The `data.DataCatalog` component scans input files, recognises each by its **content** rather than its name, records its time coverage, satellite, product class, and a **SHA-256 checksum** (a short fingerprint of the file's bytes, so each result records exactly which inputs produced it), and selects the appropriate files for a given cut-off.

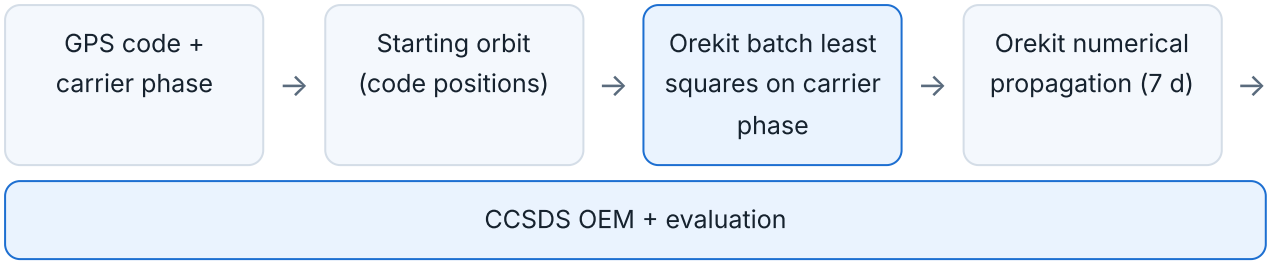
Layer 3 — the interface. A small local web server (`app.gui.GuiServer`, built on the JDK's built-in HTTP server) binds to 127.0.0.1 only and serves a browser interface. Progress is streamed to the browser; each run is stored under a run directory containing its configuration, outputs, and log, making every result reproducible. The browser is merely the display; no data leaves the computer, and even the help search operates offline.

Processing modes. A run operates in one of two product modes — **Forecast** (cut-off-safe: only products available at the cut-off are used, as in a real operation) or **Reconstruction**

(retrospective: the most accurate products present are used) — and in one of two workflows: the full **orbit-determination + forecast**, or **propagation from a single supplied state**.

11. Processing pipeline

For a chosen cut-off, the system executes the following chain on the **Orekit** flight-dynamics engine (illustrated below):



1. **Input selection** from the catalogue: the satellite's GPS observations over the arc (twelve hours by default for Sentinel-3A), the best available GPS products, the GPS antenna file, and the POEORB reference (for evaluation only). Orekit's parsers read the RINEX observations, the SP3 and clock products and the broadcast navigation messages.
2. **Starting orbit.** Single-point code positions every 300 s (§9.1) give an initial orbit through Orekit's Gibbs method and a quick Orekit batch fit. They only start the next step.
3. **Orbit determination.** An Orekit batch least-squares fit uses the undifferenced ionosphere-free **code and carrier phase** of every GPS satellite in view, every 60 s. It estimates:
   - the orbit and the drag coefficient;
   - one receiver clock per epoch;
   - one float ambiguity per continuous phase arc.

   It also handles cycle slips and the antenna offset (§9.3, CARRIER_PHASE_OD.md).
4. **Propagation.** The fitted orbit is integrated forward by the Orekit numerical propagator (seven days by default), applying the forces of Part IV, with a position recorded every 60 s (§9.2).
5. **Output.** The run writes:
   - a CCSDS OEM (optionally with the covariance from Orekit's state-transition matrix);
   - KML and CZML files;
   - a printable report;
   - a log.

Where a POEORB reference is present, the error against it is computed for display only.

> Earlier versions of this system fitted the orbit to the single-point position fixes, with its own physics library; the campaign of §13 was produced that way. §13a compares that method with the carrier-phase orbit determination on the same 61 days.

12. Features and interface

An error figure is available only for days whose Sentinel-3A reference orbit is in the bundled data; the reference is used for evaluation, never as an input.

Beyond the campaign, more than 120 automated tests verify the underlying mathematics: loss-free orbit conversions, energy conservation under two-body motion, the J2 nodal-regression formula, the leap-second table, NRLMSISE-00 against its official reference case to seven significant figures, batch-OD recovery of a known state and drag coefficient from synthetic data, and CCSDS OEM round-trips. The propagator is additionally checked against real Sentinel-3A precise orbits.

13a. Rebuild: carrier-phase orbit determination on the Orekit engine

The orbit determination was rebuilt and rerun on the same 61 cut-offs with the same settings:

- **Measurements.** It now uses the GPS **carrier phase**, not only the code.
- **Engine.** Every step now runs on **Orekit**: file parsing, frames, force models, the batch least-squares estimator and propagation.

Run	Days	Successful	Median fit RMS	Median run time
Before: SPP position fit (previous engine)	61	61	2.702 m (positions)	135 s
After: carrier-phase OD (in-house engine)	61	61	0.069 m (carrier phase)	191 s
Orekit: carrier-phase OD on the Orekit engine	61	61	0.072 m (carrier phase)	280 s

3-D position error against POEORB (m), median over the scored days (95th percentile in brackets):

Horizon	Before	After	Orekit	Change (Orekit vs before)
At cut-off	1.57 (3.21)	0.78 (2.15)	0.89 (2.22)	-44 %
+1 h	1.93 (4.48)	1.26 (4.52)	1.43 (4.68)	-26 %
+6 h	9.62 (712)	8.80 (713)	8.61 (714)	-10 %
+1 day	94.91 (4,211)	92.74 (4,212)	93.06 (4,213)	-2 %
+3 days	1,023 (16,355)	1,020 (16,361)	1,000 (16,404)	-2 %
+7 days	8,208 (42,008)	8,205 (42,014)	8,131 (42,118)	-1 %

- **At the cut-off and in the next hours,** carrier phase roughly halves the error.

- **The forecast beyond a few hours** is dominated by atmospheric drag, so it changes very little. Improving it further means better density forecasting, not a better orbit determination.

Details: CARRIER_PHASE_OD.md and reports/two-month-carrier-phase-orekit.

13b. Five further improvements

After the rebuild, five improvements were made and measured. Each was first tested on seven cut-offs; all were then run together in a second 61-day campaign.

1. **The drag coefficient is estimated, on a 12-hour arc.**
2. **Alternative drag models** were tested: JB2008, and a box-wing spacecraft model with measured attitude.
3. **GPS antenna phase centres** from the IGS ANTEX file, and **phase wind-up**, are modelled.
4. **The attitude law and output frames** now come from Orekit. The nominal attitude (yaw-compensated geodetic nadir pointing) matches the measured attitude to 0.005°.
5. **Window runs** compute several cut-offs **in parallel**.

Step 2's two models made the forecasts worse and were not adopted. The others are now the defaults.

Horizon	Before (SPP fit)	Carrier phase on Orekit	Final (steps 1–5)	Change vs before	Change vs Orekit rebuild
At cut-off	1.57 (3.21)	0.89 (2.22)	**1.77** (3.19)	+12 %	+100 %
+1 h	1.93 (4.48)	1.43 (4.68)	**2.10** (4.06)	+9 %	+47 %
+6 h	9.62 (712)	8.61 (714)	**5.57** (723)	-42 %	-35 %
+1 day	94.91 (4,211)	93.06 (4,213)	**42.79** (4,286)	-55 %	-54 %
+3 days	1,023 (16,355)	1,000 (16,404)	**445** (17,245)	-57 %	-56 %
+7 days	8,208 (42,008)	8,131 (42,118)	**10,416** (47,805)	+27 %	+28 %

Final run: 61 cut-offs, 61 successful; median phase residual 12.2 cm; median run time 732 s per cut-off (four in parallel).

- **Forecasts from 6 hours to 3 days improve by about half.** Estimating the drag coefficient absorbs most of the density-model error. The median error at +1 day falls from 95 m to 43 m, and at +3 days from 1.0 km to 0.45 km.

- **Positions near the cut-off are slightly less accurate,** about 1.8 m. A 12-hour arc fitted with the physical force model alone cannot follow the orbit as tightly at its end.
- **At seven days the median error is 27 % larger.** A drag coefficient fitted to half a day of atmosphere does not hold for a week near solar maximum. For seven-day products the earlier settings remain preferable.

Two cut-offs did not converge with the drag coefficient free; the system now refits them with it fixed. The test-day comparison, the daily values and the data are in reports/two-month-improvements.

14. Accuracy as a function of GPS product

To quantify how the choice of GPS product affects accuracy, the same case (cut-off 2024-10-15, Reconstruction mode) was run with each product tier. Two comparisons are informative.

(a) Instantaneous orbit quality (post-fit RMS at the cut-off).

Product tier	Post-fit RMS	Relative to broadcast
Broadcast	2.07 m	baseline
Ultra-rapid (CODE)	2.56 m	+0.49 m
Rapid (CODE)	2.49 m	+0.42 m
Final (CODE)	2.45 m	+0.38 m

(b) Orbit difference relative to the broadcast run, over the seven-day forecast (how far each tier's trajectory diverges from the broadcast trajectory):

Product tier	Maximum 3-D difference	RMS 3-D difference	Difference at seven days
Ultra-rapid	3.03 m	1.65 m	2.63 m
Rapid	4.86 m	2.69 m	4.37 m
Final	4.90 m	2.72 m	4.44 m

Interpretation. The tiers differ at only the **few-metre** level. When each run's error is plotted against the POEORB reference (Figure 7), the four curves very nearly coincide, because over a seven-day forecast the trajectory drifts by approximately **20 km** through atmospheric-drag uncertainty (§8), which dwarfs the metre-level differences between GPS products. The practical conclusion is significant: **for a multi-day LEO forecast, the forecast horizon and the drag model dominate the error, not the choice of GPS product.** The product choice matters most for the *instantaneous* orbit at the cut-off, where the error is at

and each force's contribution can be examined.
- **Standards-based output.** CCSDS OEM, KML and CZML allow results to be consumed directly by other space tools and viewers.
- **Reproducibility and auditability.** Each run stores its configuration and the checksums of its inputs; the engine is plain Java with no hidden dependencies.
- **Offline operation.** The system runs with no internet connection, as secure facilities require; even the help search is offline.
- **Honest validation.** Results are measured against real ESA precise orbits, which are never fed back into the computation.
- **Usability and explainability.** The interactive interface, comparisons, and illustrated guide make the system understandable rather than opaque.

17. Limitations

- **Real-time product availability.** A genuine forecast at the cut-off can use only broadcast navigation and the predicted portion of ultra-rapid products; Final and Rapid products become available one to two weeks later and are used in Reconstruction mode.
- **Forecast drift.** Forecast error grows chiefly through atmospheric drag, which depends on unpredictable solar activity. Estimating the drag coefficient (§13b) halves it over one to three days, but several kilometres over seven days remain near solar maximum without forecast space-weather input.
- **Strictly dynamic orbit.** The orbit is fitted with the physical force model only, without empirical accelerations, and the ambiguities are not fixed to integers. Phase residuals of about 6 cm remain, against the 1–2 cm of operational precise orbit determination.
- **Ocean tides** use the eight main FES2004 constituents to degree 4. The receiving antenna's phase-centre variations are not modelled.
- **Rendering.** Media were produced on hardware without a graphics processor, so the satellite model is enlarged and some captures are time-lapsed; on ordinary hardware the globe animates smoothly.
- **Licensing.** The NRLMSIS 2.0 and DTM2020 atmosphere models are not included for licensing reasons; the shipped drag options are exponential, Harris–Priester, NRLMSISE-00, and JB2008.

18. Running the software

Complete installation and execution instructions for both Linux and Windows — from installing Java and Maven through building and launching — are provided in the project **README**. In summary:

```
        mvn clean package                                  # build (add -
DskipTests to build faster)
```

```
        java -jar target/leo-trajectory-0.1.0.jar gui   # start the browser
interface
```

The program prints a local address of the form `http://127.0.0.1:…`, which opens automatically. On the Run tab, selecting **Run** (or **Compare product tiers**) computes a case; the Results and Window tabs present the output. The bundled two-month Sentinel-3A dataset is available immediately.

A window run computes several cut-offs at the same time, each in its own process: by default, one per spare processor core, limited by memory. A saved run configuration can also be executed without the interface:

```
        java -jar target/leo-trajectory-0.1.0.jar run <config.json> --out
results
```

The JPL DE440 ephemeris and the GPS antenna file are taken from `data/ephemeris/` and `data/antenna/`.

Glossary

Ordered broadly from general to specific. Every physical and mathematical term used in this report appears here.

Satellite — an object placed in orbit around the Earth.

Orbit — the path a satellite follows; physically, continuous free-fall that keeps missing the Earth.

Low Earth Orbit (LEO) — an orbit a few hundred kilometres above the surface (Sentinel-3A ≈ 815 km), with a period of about 100 minutes.

Gravity / Newton's law of gravitation — the attraction between masses, $F = GMm/r^2$; it weakens with the square of distance.

Newton's second law — force equals mass times acceleration, $F = ma$.

Acceleration — the rate of change of velocity; the second time derivative of position, $\ddot{\mathbf{r}}$.

Gravitational parameter (μ) — the product $G \times M_{\text{Earth}}$.

State vector — six numbers giving a satellite's position and velocity at an instant.

Orbital (Keplerian) elements — six numbers describing an orbit's shape and orientation: semi-major axis, eccentricity, inclination, right ascension of the ascending node, argument of perigee, and true anomaly.

Semi-major axis (a) — half the long diameter of the orbital ellipse; its size.

Eccentricity (e) — how elongated the orbit is (0 = circle).

Inclination (i) — the tilt of the orbital plane relative to the equator.

Perigee / apogee — the lowest / highest points of an orbit.

Sun-synchronous orbit — an orbit whose plane precesses about 1° per day so the satellite crosses a given latitude at a fixed local solar time.

Epoch — a specific instant at which a state or measurement is defined.

Ephemeris — a table of a satellite's position (and velocity) over time.

TAI / UTC / GPS time / UT1 — time scales: uniform atomic time; civil time (TAI minus leap seconds); GPS system time (TAI − 19 s, no leap seconds); and time tied to the Earth's rotation angle.

Ground track — the path of the point on the Earth's surface directly beneath the satellite.

SHA-256 checksum — a short fingerprint of a file's contents, recorded for reproducibility.

Air-gapped / offline — a computer with no internet connection; the system runs entirely locally.

Retrieval-augmented generation (RAG) — answering a question by first retrieving relevant documentation passages, then generating an answer from those passages with citations. **BM25** is the text-ranking method used for retrieval.

Manoeuvre flag — a warning raised when orbit-determination residuals jump, possibly indicating a thruster firing.

Carrier phase — the measured count of cycles of the GPS radio wave (19 and 24 cm wavelengths on L1 and L2), about a hundred times less noisy than the code; it carries an unknown whole number of cycles, the ambiguity.

Ambiguity — the unknown whole number of cycles in a carrier-phase measurement; constant while the receiver keeps lock, so one value is estimated per continuous phase arc.

Cycle slip — a sudden jump of a whole number of cycles in the carrier phase when the receiver briefly loses lock; it starts a new ambiguity.

Ionosphere-free combination — a weighted difference of the L1 and L2 measurements that removes the first-order delay of the signal in the ionosphere.

Phase centre / ANTEX — the point of an antenna to which its measurements refer; the IGS ANTEX file gives each GPS satellite antenna's offset from the satellite's centre of mass and its variations with direction.

Phase wind-up — the change in measured carrier phase caused by the transmitting and receiving antennas rotating relative to each other (the signal is circularly polarised).

Drag coefficient (Cd) — the dimensionless factor scaling the drag force for a given area and air density; estimating it absorbs part of the density-model error.

Orekit — an open-source flight-dynamics library (orbits, frames, time, force models, propagation, orbit determination), used here as the engine of every run.

Project: a reference-quality, extensible system for LEO orbit determination and trajectory prediction from a satellite's own GPS data. Test satellite: ESA Sentinel-3A. Validation period: 2 September – 2 November 2024.